Sake Table Standards Memo

TO: Dr. Alicia Modenbach and Dr. Michael Peterson, BAE 402 Instructors

FROM: Benjamin Lin, BAE Student

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SUBJECT: Summary of Four Standards for Sake Table Design

The purpose of this memo is to identify four industry standards applicable to the smart sake table project. The standards will guide design decisions for material selection, electronics selection, system reliability, and product bioprocessing. They ensure that the system will withstand environmental stresses while maintaining food safety and functionality.

RELEVANT STANDARDS

Thin Layer Drying of Agricultural Crops - American Society of Agricultural and Biological Engineers, 2022
Provides procedures for modeling drying characteristics of grains and other biomaterials. System calibration, component selection, and validation of the smart sake table's humidity and temperature control systems will reference this standard during design and testing phases. This will support consistent growth, drying, and temperature conditions throughout the process.

Food Equipment Materials - NSF International, 2018
Defines sanitary and toxicity guidelines for choosing materials with direct food contact in processing equipment. This standard specifically shows selection for choice of polymer, metal, wood, tubing, and gasket which are expected for the electrical and physical applications of the project. This will be referenced during the design and building phase of the project.

Environment Testing: General and Guide - International Electrotechnical Commission, 2013
Provides general framework for choosing electrical components for extensive environmental conditions like humidity, temperature, and vibrations. This standard applies to the electrical components of the project, ensuring design validation and reliability in the humid, hot environments. This will be referenced during the testing and building phases of the project.

Density, Specific Gravity, and Mass-Moisture Relationships of Grain Storage - American Society of Agricultural Engineers, 1992
Describes density, specific gravity, and moisture content for different stored grains. This standard will inform the drying and temperature control process in rice processing. It will apply to the design and testing phases of the project, as well as additional modeling information for a COMSOL proof of concept.

Integrating these standards will support safe, reliable, and consistent operation of the smart sake table. They provide a foundation for following food-grade standards, ensuring mechanical and electrical performance, and informed component selection. Thank you for reviewing this memo.

REFERENCES

American Society of Agricultural and Biological Engineers. (2022). *ANSI/ASAE S448.2 SEP2014 (R2022): Thin-Layer Drying of Agricultural Crops*. https://elibrary.asabe.org/abstract.asp?aid=49062

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